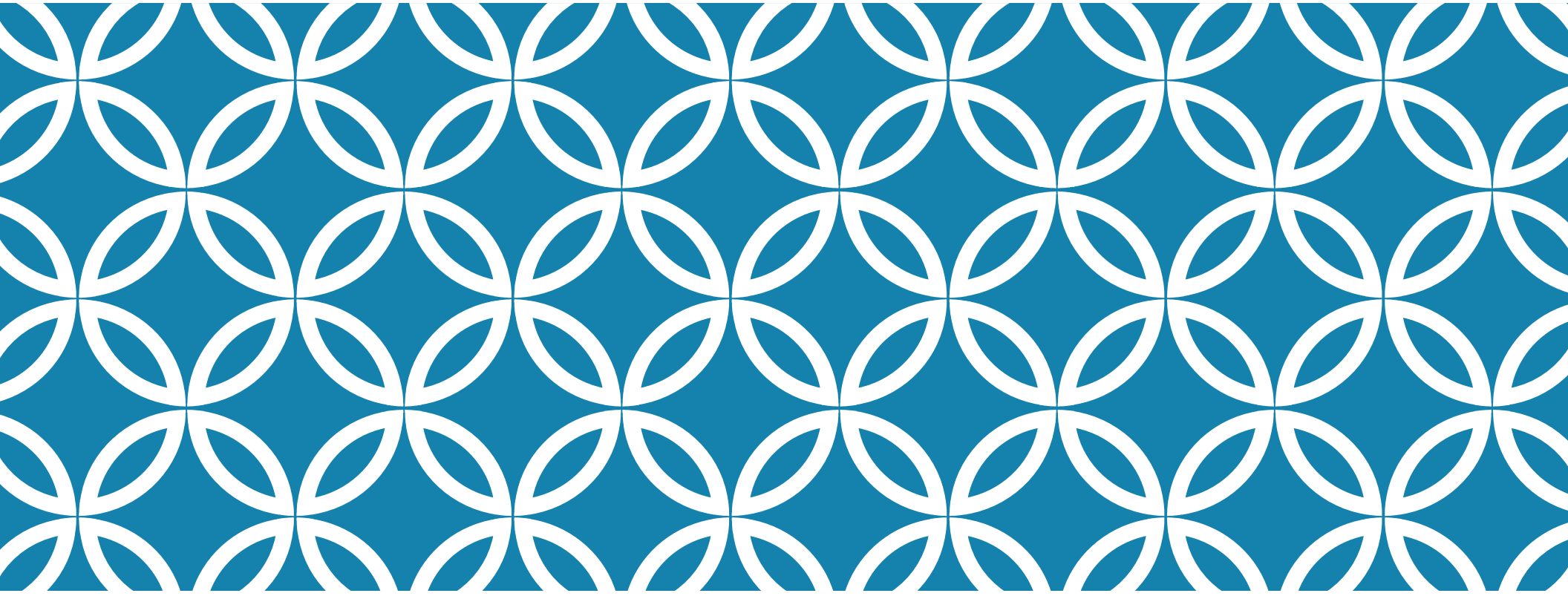




LECTURE 2

Digital Logic Design

OCTAL NUMBER SYSTEM

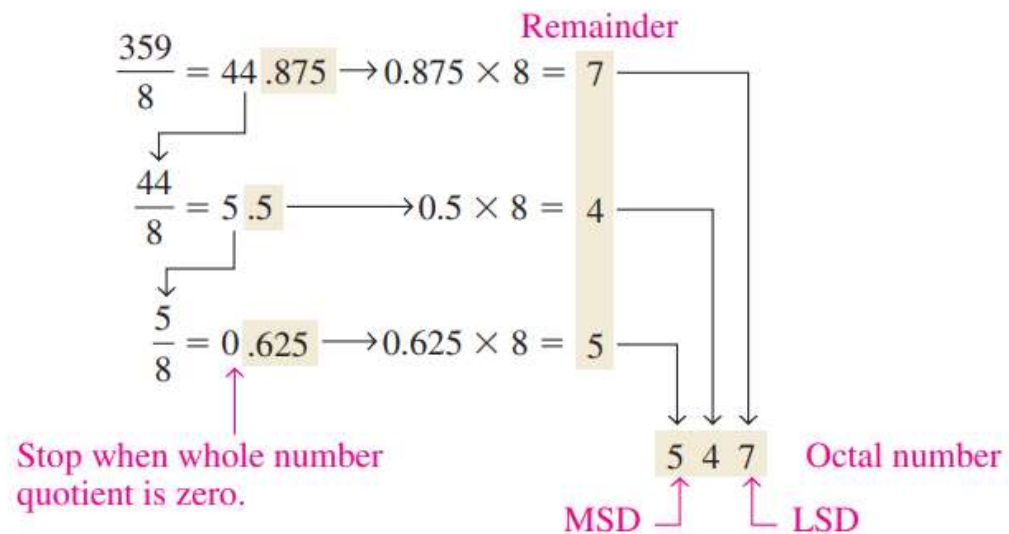
The octal number system is composed of eight digits, which are;

0, 1, 2, 3, 4, 5, 6, 7

OCTAL-TO-DECIMAL CONVERSION

$$\begin{aligned}(2374)_8 &= (2 \times 8^3) + (3 \times 8^2) + (7 \times 8^1) + (4 \times 8^0) \\&= (2 \times 512) + (3 \times 64) + (7 \times 8) + (4 \times 1) \\&= 1024 + 196 + 56 + 4 \\&= 1276_{10}\end{aligned}$$

DECIMAL-TO-OCTAL CONVERSION



OCTAL-TO-BINARY CONVERSION

Because each octal digit can be represented by a 3-bit binary number, it is very easy to convert from octal to binary.

BINARY-TO-OCTAL CONVERSION

Convert binary into octal.

110011011

Solution:

If we break number into 3 bits.

So;

110	6
011	3
011	3

$(110011011)_2 = (633)_8$

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

HEXADECIMAL NUMBERS

The hexadecimal number system has sixteen characters; it is used primarily as a compact way of displaying or writing binary numbers because it is very easy to convert between binary and hexadecimal. As you are probably aware, long binary numbers are difficult to read and write because it is easy to drop or transpose a bit.

Machine language uses hexadecimal number system.

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

BINARY-TO-HEXADECIMAL CONVERSION

Converting a binary number to hexadecimal is a straightforward procedure. Simply break the binary number into 4-bit groups, starting at the right-most bit and replace each 4-bit group with the equivalent hexadecimal symbol.

Convert the following binary numbers to hexadecimal:

1100101001010111

Solution:

1100 1010 0101 0111

Now breaking in four bit group;

1100	C
1010	A
0101	5
0111	7

Therefore;

$$(1100101001010111)_2 = (CA57)_{16}$$

Convert the following binary numbers;

111111000101101001

HEXADECIMAL-TO-BINARY CONVERSION

To convert from a hexadecimal number to a binary number, reverse the process and replace each hexadecimal symbol with the appropriate four bits.

Determine the binary numbers from hexadecimal numbers;

10A4

1	0001
0	0000
A	1010
4	0100

$$(10A4)_{16} = (0001000010100100)_2$$

HEXADECIMAL-TO-DECIMAL CONVERSION

One way to find the decimal equivalent of a hexadecimal number is to first convert the hexadecimal number to binary and then convert from binary to decimal.

Convert the following hexadecimal numbers to decimal:

- (a) $1C_{16}$ (b) $A85_{16}$

Solution

Remember, convert the hexadecimal number to binary first, then to decimal.

(a)

$$\begin{array}{c} 1 \quad C \\ \downarrow \quad \downarrow \\ \overbrace{0001} \overbrace{1100} = 2^4 + 2^3 + 2^2 = 16 + 8 + 4 = \mathbf{28}_{10} \end{array}$$

(b)

$$\begin{array}{c} A \quad 8 \quad 5 \\ \downarrow \quad \downarrow \quad \downarrow \\ \overbrace{1010} \overbrace{1000} \overbrace{0101} = 2^{11} + 2^9 + 2^7 + 2^2 + 2^0 = 2048 + 512 + 128 + 4 + 1 = \mathbf{2693}_{10} \end{array}$$

Related Problem

Convert the hexadecimal number 6BD to decimal.

E5₁₆ to Decimal:

E in hexadecimal represents 14 in decimal.

5 in hexadecimal represents 5 in decimal.

The hexadecimal number E5₁₆ can be broken down as:

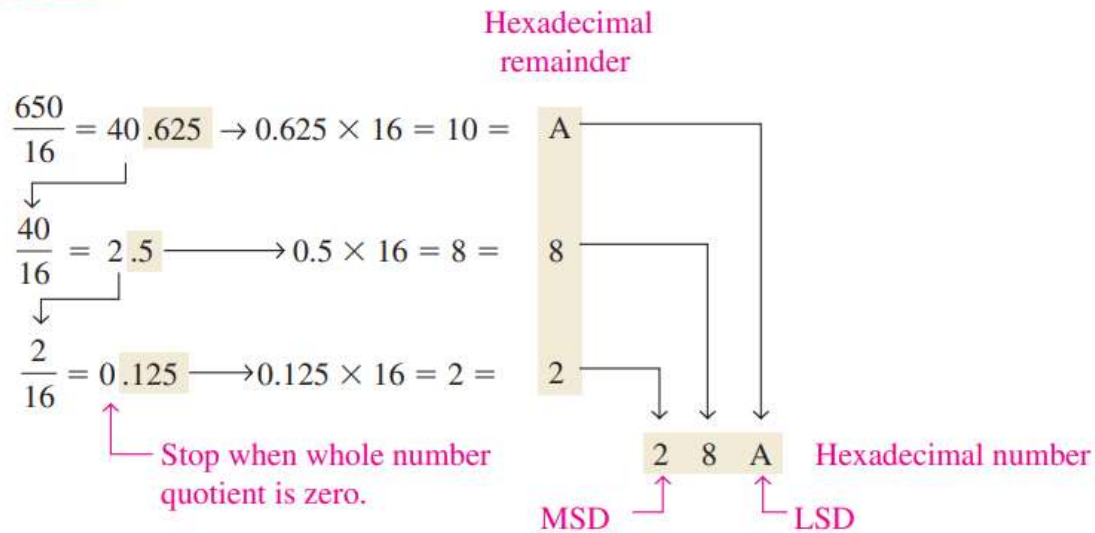
$$(14 * 16^1) + (5 * 16^0)$$

This equals: $(14 * 16) + (5 * 1) = 224 + 5 = 229_{10}$

DECIMAL-TO-HEXADECIMAL CONVERSION

Convert the decimal number 650 to hexadecimal by repeated division by 16.

Solution



Convert each of the following octal numbers to binary:

(a) 13_8 (b) 25_8 (c) 140_8 (d) 7526_8

Solution

(a) 1 3
 ↓ ↓
 └─┘ └─┘
 001011

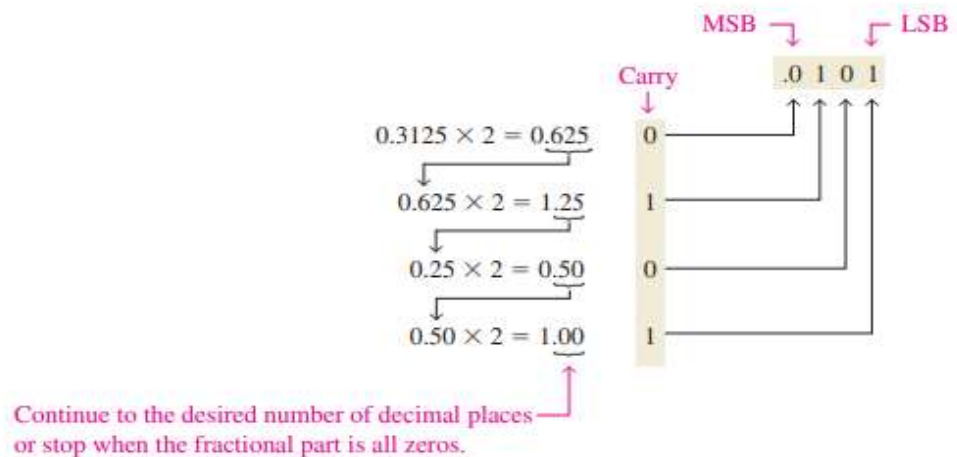
(b) 2 5
 ↓ ↓
 └─┘ └─┘
 010101

(c) 1 4 0
 ↓ ↓ ↓
 └─┘ └─┘ └─┘
 001100000

(d) 7 5 2 6
 ↓ ↓ ↓ ↓
 └─┘ └─┘ └─┘ └─┘
 111101010110

DECIMAL FRACTION TO BINARY

Repeated Multiplication by 2 As you have seen, decimal whole numbers can be converted to binary by repeated division by 2. Decimal fractions can be converted to binary by repeated multiplication by 2. For example; 0.3125 [Practice question 0.375]



BINARY ARITHMETIC

Addition:

Four basic rules of binary addition;

$$0 + 0 = 0$$

Sum of 0 with carry of 0

$$0 + 1 = 1$$

Sum of 1 with carry of 0

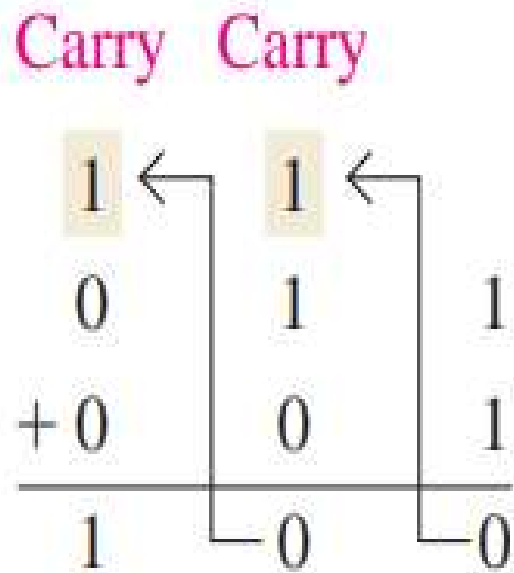
$$1 + 0 = 1$$

Sum of 1 with carry of 0

$$1 + 1 = 10$$

Sum of 0 with carry of 1

EXAMPLE



```
  1101
+ 1010
-----
10111
```

Subtraction:

$$0 - 0 = 0$$

$$1 - 1 = 0$$

$$1 - 0 = 1$$

$$10 - 1 = 1$$

[0-1 with a borrow of 1]

Let's examine exactly what was done to subtract the two binary numbers since a borrow is required. Begin with the right column.

Left column:

When a 1 is borrowed,
a 0 is left, so $0 - 0 = 0$.

Middle column:

Borrow 1 from next column
to the left, making a 10 in
this column, then $10 - 1 = 1$.

Right column:

$1 - 1 = 0$

$$\begin{array}{r} 0101 \\ -011 \\ \hline 010 \end{array}$$

Multiplication:

Four basic rules of multiplication are;

$$0 \times 0 = 0$$

$$0 \times 1 = 0$$

$$1 \times 0 = 0$$

$$1 \times 1 = 1$$

Perform the following binary multiplications:

(a) 11×11

(b) 101×111

Solution

(a)

$$\begin{array}{r} 11 \qquad 3 \\ \times 11 \quad \times 3 \\ \hline 11 \qquad 9 \\ +11 \qquad \\ \hline 1001 \end{array}$$

Partial products {

(b)

$$\begin{array}{r} 111 \qquad 7 \\ \times 101 \quad \times 5 \\ \hline 111 \qquad 35 \\ 000 \\ +111 \\ \hline 100011 \end{array}$$

Partial products {

Division of Binary numbers:

It is the same as division of decimal number.

$$\begin{array}{r} 1001 \\ 11 \overline{) 11011} \\ \underline{-11} \\ 00 \\ \underline{-01} \\ 1 \\ \underline{-0} \\ 11 \\ \underline{-11} \\ 0 \end{array}$$

$$(11011)_2 \div (11)_2 = (1001)_2$$

COMPLEMENTS OF BINARY NUMBER

1's complement:

Finding the 1's Complement The 1's complement of a binary number is found by changing all 1s to 0s and all 0s to 1s, as illustrated below:

1 0 1 1 0 0 1 0	Binary number
↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓	
0 1 0 0 1 1 0 1	1's complement

2's Complement:

The 2's complement of a binary number is found by adding 1 to the LSB of the 1's complement.

$$2's\ complement = 1's\ complement + 1$$

Find 2's complement of 0000 0101

Step 1:

Take One's complement of binary number:

Write the binary Number

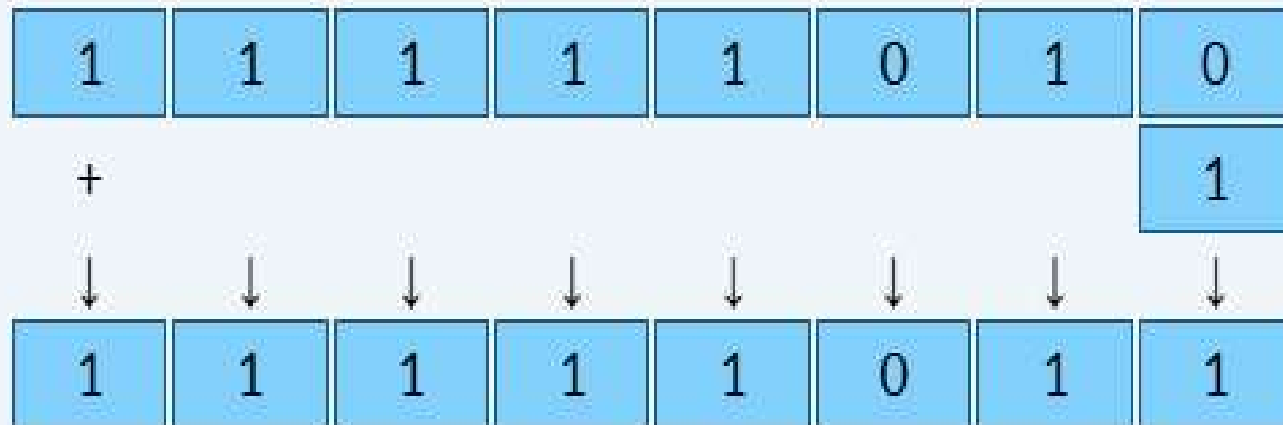
0	0	0	0	0	1	0	1
---	---	---	---	---	---	---	---

Reverse all the values (Interchange each 0 with 1 and 1 with 0):

0	0	0	0	0	1	0	1
↓	↓	↓	↓	↓	↓	↓	↓
1	1	1	1	1	0	1	0

Step 2:

Take Two's complement by adding 1 in previously written binary number:



SIGNED BITS

Digital systems, such as the computer, must be able to handle both positive and negative numbers. A signed binary number consists of both sign and magnitude information. The sign indicates whether a number is positive or negative, and the magnitude is the value of the number.

There are three forms in which signed integer (whole) numbers can be represented in binary: sign-magnitude, 1's complement, and 2's complement.

Of these, the 2's complement is the most important and the sign-magnitude is the least used.

Non integer and very large or small numbers can be expressed in floating-point format.

THE SIGN BIT

The left-most bit in a signed binary number is the sign bit, which tells you whether the number is positive or negative.

A 0 sign bit indicates a positive number, and a 1 sign bit indicates a negative number.

00011001
Sign bit ——— ↑ ↑ ——— Magnitude bits

The decimal number -25 is expressed as

10011001

Notice that the only difference between $+25$ and -25 is the sign bit because the magnitude bits are in true binary for both positive and negative numbers.

Express the decimal number -39 as an 8-bit number in the sign-magnitude, 1's complement, and 2's complement forms.

Solution

First, write the 8-bit number for $+39$.

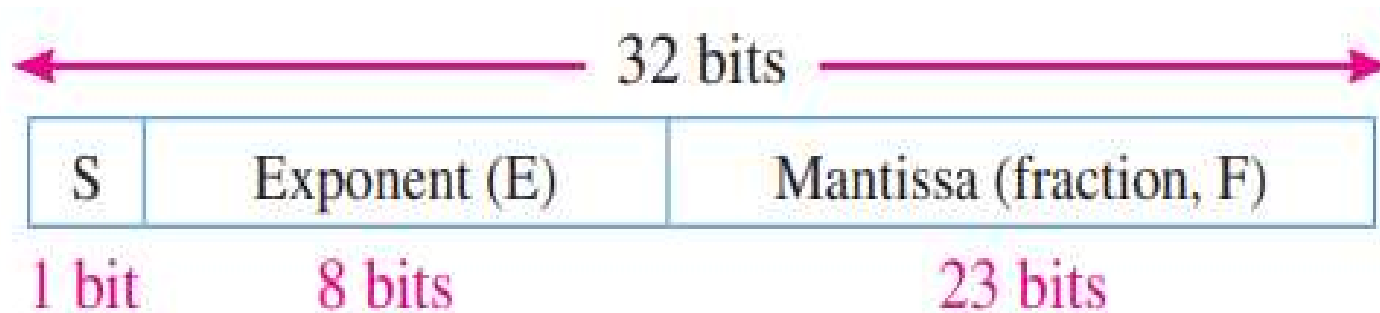
00100111

In the *sign-magnitude form*, -39 is produced by changing the sign bit to a 1 and leaving the magnitude bits as they are. The number is

10100111

FLOATING-POINT NUMBERS

A floating-point number (also known as a real number) consists of two parts plus a sign. The mantissa is the part of a floating-point number that represents the magnitude of the number and is between 0 and 1. The exponent is the part of a floating-point number that represents the number of places that the decimal point (or binary point) is to be moved.



Illustrate binary number 1011010010001 as floating point number.

Solution:

Step 1:

First, it can be expressed as 1 plus a fractional binary number by moving the binary point 12 places to the left and then multiplying by the appropriate power of two.

$$1011010010001 = 1.011010010001 \times 2^{12}$$

Step 2:

Assuming that this is a positive number, the sign bit (S) is 0. The exponent, 12, is expressed as a biased exponent by adding it to 127 ($12 + 127 = 139$). The biased exponent (E) is expressed as the binary number 10001011.

Step 3:

The mantissa is the fractional part (F) of the binary number, .011010010001. Because there is always a 1 to the left of the binary point. in the power-of-two expression, it is not included in the mantissa. The complete floating point number is:

S	E	F
0	10001011	011010010001000000000000

When you have calculated these binary values, you can put them into a 32- or 64-bit field. The digits are arranged like this:

Single precision (32-bit) form: (Bias = 127)



Double precision (64-bit) form: (Bias = 1023)



(The numbers in parentheses show how many bits are required in each field.)

BINARY CODED DECIMAL

Binary coded decimal (BCD) is a way to express each of the decimal digits with a binary code. There are only ten code groups in the BCD system, so it is very easy to convert between decimal and BCD. Because we like to read and write in decimal, the BCD code provides an excellent interface to binary systems.

DECIMAL TO BCD

Decimal	BCD
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

DECIMAL TO BCD

Convert each of the following decimal numbers to BCD:

- (a) 35 (b) 98 (c) 170 (d) 2469

Solution

(a) 3 5
 ↓ ↓
 00110101

(b) 9 8
 ↓ ↓
 10011000

(c) 1 7 0
 ↓ ↓ ↓
 000101110000

(d) 2 4 6 9
 ↓ ↓ ↓ ↓
 0010010001101001

BCD TO DECIMAL

Convert each of the following BCD codes to decimal:

- (a) 10000110 (b) 001101010001 (c) 1001010001110000

Solution

- (a) $\overbrace{1000}^{8}\overbrace{0110}^{6}$ (b) $\overbrace{0011}^{3}\overbrace{0101}^{5}\overbrace{0001}^{1}$ (c) $\overbrace{1001}^{9}\overbrace{0100}^{4}\overbrace{0111}^{7}\overbrace{0000}^{0}$

GRAY CODE

The Gray code is unweighted and is not an arithmetic code; that is, there are no specific weights assigned to the bit positions. The important feature of the Gray code is that it exhibits only a single bit change from one code word to the next in sequence.

TABLE 2-6

Four-bit Gray code.

Decimal	Binary	Gray Code	Decimal	Binary	Gray Code
0	0000	0000	8	1000	1100
1	0001	0001	9	1001	1101
2	0010	0011	10	1010	1111
3	0011	0010	11	1011	1110
4	0100	0110	12	1100	1010
5	0101	0111	13	1101	1011
6	0110	0101	14	1110	1001
7	0111	0100	15	1111	1000

GRAY CODE

- (a) Convert the binary number 11000110 to Gray code.
(b) Convert the Gray code 10101111 to binary.

Solution

- (a) Binary to Gray code:



- (b) Gray code to binary:



Related Problem

- (a) Convert binary 101101 to Gray code.
(b) Convert Gray code 100111 to binary.